

SMRUTISHREE BEHERA

PhD Researcher in Physics | Ferroelectric Materials & Neuromorphic Devices

SRM University-AP, India

+91 9861623218 | smrutishreebehera2001@gmail.com

RESEARCH PROFILE

Physics researcher pursuing a PhD at SRM University-AP, India, working on ferroelectric materials with a focus on their potential for neuromorphic device applications. Academic background in condensed matter physics, solid-state physics and quantum mechanics, with experience in experimental characterization, electronics and scientific software. Developing research interests include ferroelectric-enhanced neurotechnology, neural interfaces and neural modulation.

RESEARCH INTERESTS

Ferroelectric materials and functional properties; material-based memory and neuromorphic devices; ferroelectric-enhanced neurotechnology and neural modulation.

RESEARCH EXPERIENCE

PhD Research - Ferroelectric Materials

SRM University-AP, India | Ongoing

- Working on the physical and functional properties of ferroelectric materials, with an interest in their use in neuromorphic hardware and memory devices.
- Exploring the potential connection between functional ferroelectric materials, neurotechnology and neural modulation.

Study of the Quark Model and Classification of Hadrons

Master's project | Kendrapara Autonomous College, affiliated to Utkal University | 3 months

- Explored SU(3) flavor symmetry and analyzed baryon and meson families.
- Compared theoretical predictions of hadron properties with experimental data, strengthening a foundation in quantum chromodynamics.

Design and Optimization of a Colpitts Oscillator Circuit

B.Sc. project | Berhampur University | 1 month

- Designed oscillator circuits to study frequency stability and waveform generation using transistor and op-amp setups.
- Investigated the effects of passive components on oscillation frequency and circuit performance.
- Developed practical skills in analog electronics, circuit testing and signal analysis.

SMRUTISHREE BEHERA

Academic background & technical skills

EDUCATION

PhD in Physics

SRM University-AP, India | Ongoing

Research focus: ferroelectric materials and their functional properties, with a focus on neuromorphic device applications. Developing interests: ferroelectric-enhanced neurotechnology and neural modulation.

M.Sc. in Physics

Kendrapara Autonomous College, affiliated to Utkal University | Oct 2022 - Jul 2024

CGPA: 7.35 (Grade B). **Coursework:** Advanced Condensed Matter Physics I & II, Basic Condensed Matter Physics, Advanced Quantum Mechanics, Statistical Mechanics and Dissertation/Project.

Bachelor of Science - Physics Honours

Berhampur University | Jul 2019 - Jul 2022

CGPA: 8.66 (Grade A+). **Coursework:** Solid State Physics, Quantum Mechanics & Applications, Nano Materials & Applications, Statistical Mechanics, Electromagnetic Theory, Elements of Modern Physics and Classical Dynamics.

Higher Secondary School

Council of Higher Secondary Education, Odisha | 2018

Result: 76.5%. **Subjects:** Physics, Chemistry, Mathematics and Biology (Botany & Zoology).

High School

Board of Secondary Education, Odisha | 2016

Result: 90.0%. **Subjects:** Mathematics, General Science, Social Science, English and Odia.

TECHNICAL SKILLS

Electrical characterization: Impedance analyzer, source meter.

Structural and spectroscopic characterization: XRD, FTIR, Raman.

Software and instrumentation: Origin, LabVIEW.

Programming: C, Fortran.

Electronics: Analog circuit design, circuit testing and signal analysis.

ADDITIONAL INFORMATION

Languages: English, Hindi, Odia.

Paper review: Which came first, "Galaxy or Blackhole".

Interests: Dancing, drawing, reading and listening to music.